Ontology-Grounded Semantic Entity Retrieval for Digitized Traditional Vietnamese Medicine Literature: A Chatbot-Integrated Framework

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Abstract. Extracting knowledge from Traditional Medicine (TM) remains challenging due to the unstructured nature of historical texts and the complex, dialectical relationships among herbs, formulas, and therapeutic effects. Conventional text-based Retrieval-Augmented Generation (RAG) and keyword search methods often return fragmented results, while generic Large Language Modelse (LLMs) are prone to generating unverified medical information due to hallucination. To address these limitations, this paper proposes an ontology-grounded semantic entity retrieval framework for digitized Traditional Vietnamese Medicine literature, integrated with a chatbot interaction layer. The system performs semantic restructuring by converting historical documents into an ontology-compliant knowledge graph encompassing over 700 medicinal plants. Unlike systems operating on raw textual contexts, this graph explicitly models medical entities (e.g., herbs, formulas, diseases) and their therapeutic relationships. Operating on a decoupled multi-agent architecture, the system translates natural language into executable queries, enabling precise graph-based reasoning. This approach enforces an evidence-grounded generation pipeline, strictly bounding responses to the structured semantic space to substantially mitigate hallucination risks. Evaluated using the Ragas methodology, the framework demonstrates promising semantic retrieval reliability and ontology consistency. Achieving a Faithfulness score of 0.8101 and a Context Recall of 0.8251, the system empirically proves its effectiveness in executing complex, multi-hop medical retrieval tasks compared to an advanced 1-Agent sequential baseline.

Keywords: Semantic Retrieval; Knowledge Graph; Ontology; GraphRAG; Traditional Medicine.

1 Introduction

Traditional Vietnamese Medicine (TVM) is a valuable knowledge repository that plays an important role in public healthcare through the use of natural herbs and traditional remedies

(World Health Organization, 2013). However, access to this knowledge remains limited, as most information is still stored in unstructured historical medical texts. Consequently, knowledge about medicinal ingredients is often fragmented and lacks explicit connections, making it difficult to retrieve complex relationships such as herbal combinations, pharmacological interactions, and therapeutic effects (Li et al., 2020). In practice, existing retrieval systems mainly rely on keyword-based search, which frequently produces incomplete results and fails to capture semantic relationships. Alternatively, generic Large Language Models (LLMs) enable more natural interactions but introduce the risk of hallucination, potentially generating unverified medical information (Ji et al., 2023). These limitations highlight the critical need for systems capable of preserving structured medical knowledge while supporting reliable semantic querying.

To address this gap, this paper proposes an ontology-grounded semantic medical knowledge retrieval framework for TVM, supported by a conversational interaction layer. Instead of operating directly on raw textual contexts, the system semantically restructures digitized historical medical texts. This approach explicitly represents dialectical relationships among herbs, formulas, active ingredients, and diseases within a strictly defined ontology. By integrating this structured semantic space with a controlled reasoning mechanism, the framework ensures more accurate, multi-hop, and traceable information retrieval. This study utilizes a dataset of over 700 Vietnamese medicinal plants extracted and standardized from authoritative medical literature.

The main contributions of this paper are threefold: (1) a Medallion data pipeline that transforms unstructured historical texts into an ontology-driven knowledge graph; (2) a graph-based reasoning mechanism supporting explicit multi-hop therapeutic queries; and (3) an evidence-grounded chatbot interaction layer that strictly anchors generative outputs in verified data to mitigate hallucination. Empirical results demonstrate the system's high reliability in semantic retrieval and factual consistency without relying on unverified generative assumptions. The remainder of this paper is organized into related work (Sec. 2), methodology (Sec. 3), experiments (Sec. 4), and conclusion (Sec. 5).

2 Related Work

On a global scale, the modernization of Traditional Medicine, particularly Traditional Chinese Medicine (TCM), has made significant progress through the development of digital knowledge ecosystems (Wang et al., 2021). In the broader context of natural product evaluation, machine learning models such as multi-layer perceptrons and adaptive decision trees have demonstrated significant success in assessing the quality and safety of biological consumables (Duy et al., 2023; Van-An et al., 2023; Pham et al., 2023; Ngo-Ho et al., 2024). However, while these predictive models excel at numerical quality control for ingestible items, capturing the complex dialectical therapeutic relationships of traditional medicinal herbs requires a fundamentally different structural approach, underscoring the necessity of an ontology-grounded semantic framework.Recent efforts have focused on building large-scale knowledge bases such as TCMSP (Ru et al., 2014) and SymMap (Wu et al., 2019), which integrate information ranging from medicinal compounds to molecular targets. A notable trend in this area is the transition from traditional relational databases to graph-based knowledge representation (Rotmensch et al., 2017; Hogan et al., 2021). Knowledge Graphs enable explicit modeling of complex relationships and support

semantic reasoning and multi-hop queries, which are essential for traditional medicine applications (Yu et al., 2017).

In this context, ontologies have been utilized to formalize complex pharmacological knowledge and resolve semantic ambiguities. Prominent systems, such as the Traditional Chinese Medicine Language System (TCMLS) and various TCM Ontologies, map dialectical relationships into machine-readable taxonomies to support logical clinical reasoning (Chen & Wang, 2006; Zhou et al., 2021).

However, these ontological frameworks present two notable limitations: they lack the specific indigenous taxonomies required for Traditional Vietnamese Medicine (TVM), and they primarily function as static semantic repositories relying on strict query languages (e.g., SPARQL). In parallel, recent studies have explored the integration of Knowledge Graphs with Large Language Models (LLMs) through Retrieval-Augmented Generation (RAG) to overcome these static querying limitations, improving query flexibility and reducing hallucination (Lewis et al., 2020; Ji et al., 2023; Edge et al., 2024).

In Vietnam, the digitization of medicinal knowledge has also seen progress, with contributions from institutions such as the National Institute of Medicinal Materials (2017) and community-driven platforms like Vietherb (Nguyen & Tran, 2022). These systems play an important role in preserving indigenous knowledge by providing structured descriptions, images, and usage information (Do, 2004). However, these current platforms exhibit several technical limitations. Firstly, most systems rely heavily on keyword-based search, requiring exact input of medicinal names, which is problematic due to the presence of synonyms and linguistic variations in Vietnamese (Vu et al., 2022). Secondly, the lack of semantic connectivity between data entries prevents the system from supporting complex queries such as multi-hop reasoning. Finally, relationships among herbs, formulas, and treatments are not explicitly modeled, limiting the system's ability to support automated reasoning and semantic relation disambiguation. These observations reveal a clear research gap, as existing systems either lack a rigorous semantic structure or rely on generic generative models that are susceptible to unverified outputs. Therefore, there remains a critical need for a comprehensive framework that combines ontology-driven knowledge representation with constrained semantic retrieval for Traditional Vietnamese Medicine. To address this gap, this study proposes an ontology-grounded semantic medical knowledge retrieval framework, supported by a chatbot interaction layer. Rather than simply applying natural language processing for text-to-query translation, the framework systematically restructures unstructured historical records into a highly interconnected, ontology-compliant Knowledge Graph comprising over 700 digitized medicinal plants (Do, 2004). By integrating this standardized semantic space with graph-based reasoning algorithms, the system ensures that retrieved information is strictly evidence-grounded, supporting accurate multi-hop retrieval while substantially reducing hallucination risks within evaluated settings.

3 Proposed Methodology

Unlike conventional text-based Retrieval-Augmented Generation (RAG) systems or generic conversational agents that bundle natural language processing and generation into a single Large Language Model (LLM), this study proposes an ontology-grounded, functionally decoupled multi-agent architecture. This framework is specifically designed to enforce semantic control and substantially mitigate hallucination risks. The system operates on a three-agent workflow strictly

integrated with the semantic knowledge graph (Fig. 1), functioning as a structured graph-based reasoning framework engine rather than a mere conversational tool.

First, the User Query is ingested by Agent 1: NLU Parser, which performs semantic intent analysis and extracts key medical entities mapped to the ontology. The parsed request is then routed to the Retrieval Module, where the Cypher Builder acts as a central orchestrator. To minimize hardware costs associated with dense vector computation, it implements a dynamic routing mechanism. It prioritizes Exact Match and Constraints Search directly against the Knowledge Graph database for explicitly identified entities, dynamically bypassing Semantic Matching unless contextually required. Once the graph reasoning process is executed, the retrieved raw results pass through a Result Filter. This step aims to significantly reduce background noise and standardize the retrieved entities into a highly structured therapeutic representation.

Finally, the refined semantic data enters the Generation and Safety cluster, establishing a dual-control phase crucial for system reliability. Here, Agent 2: Generator synthesizes an initial draft response grounded strictly in the retrieved graph evidence. Immediately afterward, this draft is cross-validated by Agent 3: Evaluator. By verifying generated claims against the ground-truth facts, the evaluating agent detects and filters out unverified information. The system outputs the final response only if it is closely aligned with the source data. Ultimately, this decoupled architecture ensures that generated responses are backed by verifiable evidence, empirically improving the factual consistency of the medical knowledge retrieval framework.

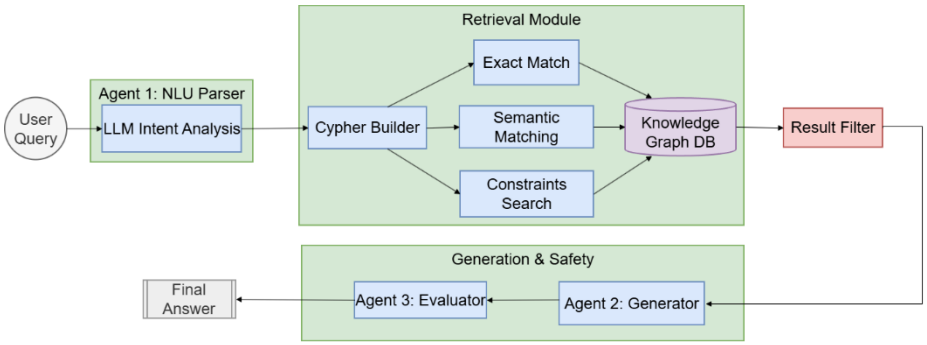


Fig 1. 3-Agent GraphRAG Architecture.

Operationally, the framework governs the reasoning process through two core algorithmic mechanisms. First, the dynamic graph query generation extracts the user's intent, entities, and constraints to populate predefined Cypher templates, ensuring precise mapping to canonical graph nodes. Subsequently, the evidence-grounded validation mechanism extracts propositional claims from the draft response and strictly cross-references them against the retrieved graph data, immediately triggering a system fallback warning if any unverified information is detected. To address the unstructured and polysemous nature of historical Traditional Medicine texts, this research implements a multi-layered semantic restructuring pipeline based on the Medallion architecture (Fig. 2). Data is sequentially refined through Bronze (raw ingestion), Silver (semantic extraction), and Diamond (ontology alignment) stages to systematically improve data integrity and semantic retrievability.

In the Bronze layer, a foundational dataset of 712 medicinal plants, extracted from Medicinal Plants and Remedies in Vietnam (Do, 2004), is digitized. To prevent context loss during

chunking, the pipeline uses semantic anchors and regular expressions to establish precise boundaries for individual herbal entries. Each segment is preserved alongside source metadata (e.g., document origins and page numbers) to guarantee evidence-grounded traceability during the generation phase.

The Silver layer transforms this unstructured text into structured representations using Large Language Models (LLMs) constrained by predefined ontology schemas. To mitigate context window limitations and reduce structural errors, the framework employs a "divide-and-conquer" strategy, modularizing the extraction into independent semantic groups (identification, formulas, ingredients, and pharmacology). This decomposition ensures strict schema adherence and facilitates isolated error correction.

Finally, the Diamond layer executes semantic normalization and validation prior to graph database ingestion. Using deterministic rule-based algorithms, this stage normalizes entity identifiers, resolves nomenclature variations via a master synonym dictionary, and audits relational links to eliminate orphan nodes or invalid connections. The result is a rigorously structured, ontology-compliant dataset optimized for explicit graph-based reasoning.

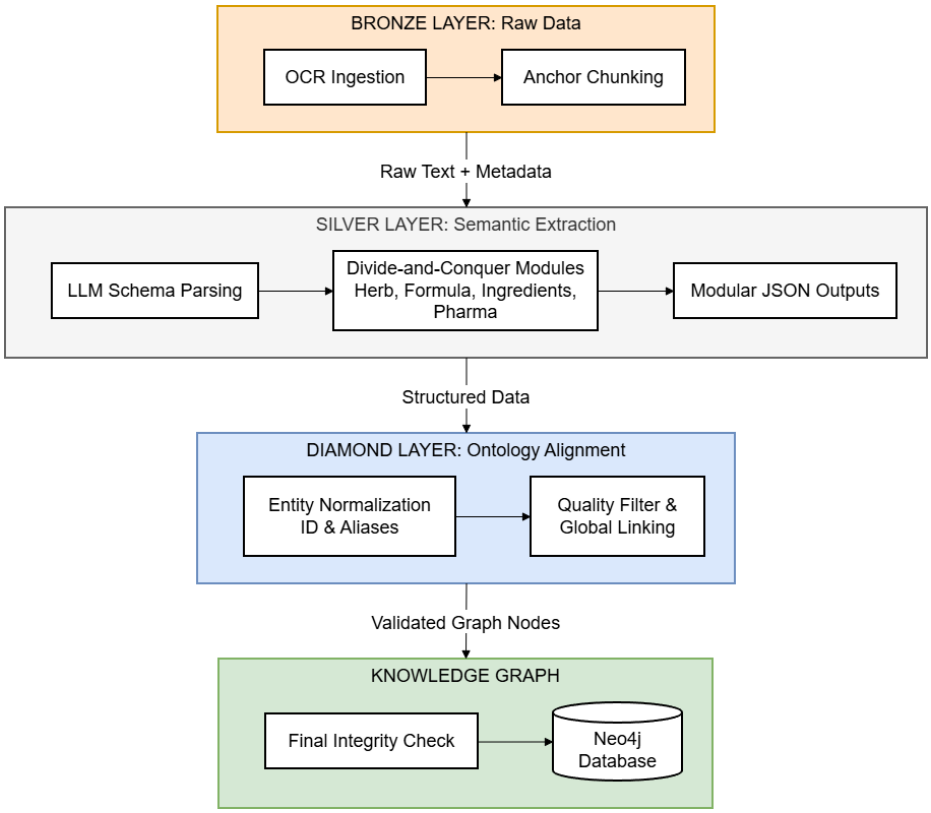


Fig 2. Professional Data Pipeline based on the Medallion Architecture.

Traditional Medicine (TM) knowledge is inherently dialectical, characterized by complex, multi-layered relationships among herbs, formulas, and symptoms. Conventional Relational Database Management Systems (RDBMS) struggle with the multi-hop queries and flexible schema

expansions required by unstructured medical texts (Noy & McGuinness, 2001). Thus, an ontology-driven representation offers fundamental advantages. First, it aligns with TM's philosophical structure (e.g., "Monarch-Minister-Assistant-Guide"), enabling semantic normalization and contextual medical logic over superficial keyword matching (Yu et al., 2017). Second, explicitly structured relationships within a Knowledge Graph efficiently support multi-hop reasoning (e.g., finding herbs with identical properties but different meridian tropisms), which is computationally expensive in SQL environments (Hogan et al., 2021). Third, the ontology acts as a rigorous semantic constraint layer. By anchoring retrieved information in validated therapeutic structures, it prioritizes factual consistency over generative assumptions, substantially mitigating hallucination risks in Large Language Models (LLMs) (Pan et al., 2024).

Applying this logic to the dataset of 712 medicinal plants, we developed a conceptual schema featuring nine primary entity nodes across three semantic layers: core entities (Herb, Formula, Disease), traditional attributes (Symptom, Property, Flavor, Meridian, Function), and modern pharmacological attributes (Active Ingredient, Pharmacology). Directed edges linking these nodes simulate TM's dialectical logic. This strict schema enforces relation disambiguation, ensuring each link carries the precise semantic weight required for explicit graph-based reasoning (Table 1).

Table 1. Relationship matrix in the TVM Knowledge Graph.

Relationship Name	Source	Target	Semantic Meaning
CONTAINS_HERB	Formula	Herb	Represents ingredients and compatibility structure
TREATS_DISEASE	Herb / Formula	Disease	Therapeutic indications for reverse querying from diseases
ALLEVIATES_SYMPTOM	Herb / Formula	Symptom	Therapeutic indications specifically for alleviating symptoms.
HAS_PROPERTY	Herb	Property	Thermal characteristics (Cold, Hot, Warm, Cool)
HAS_FLAVOR	Herb	Flavor	Taste profiles (Pungent, Bitter, Sweet, etc.)
TARGETS_MERIDIAN	Herb	Meridian	Action pathways toward internal organs
HAS_FUNCTION	Herb / Formula	Function	Effects based on TM theoretical reasoning
CONTAINS_INGREDIENT	Herb	Ingredient	Chemical composition
HAS_PHARMACOLOGY	Herb	Pharmacology	Biological effects according to modern medicine
CONTRAINDICATED_FOR	Herb / Formula	Patient / Disease	Contraindications and safety warnings.

To mitigate ambiguity among similarly named entities, the framework represents each node using an expanded semantic context rather than isolated keywords. Specifically, the entity's canonical name, aliases, curated search hints, and structural relationships are synthesized into a comprehensive descriptive string. By projecting these strings into a semantically rich vector space, the system enables similarity searches based on contextual patterns rather than simple character sequences. This approach substantially improves entity identification across synonymous or non-standardized nomenclature, ensuring strict alignment with the medical ontology (Sun et al., 2019).

For interaction, the framework implements a Semantic Parsing and Graph-Based Reasoning layer. Utilizing few-shot prompting, the Semantic Analysis module maps natural language into precise constraints by extracting three core components: the intent, the primary entity, and the constraint entity. To optimize computational efficiency, recurring query results are cached locally.

The transformation into executable graph queries follows a strictly structured sequence. First, text undergoes NFKD normalization (e.g., “Ích mẫu” to ICH_MAU, en. “Leonurus japonicus”) to ensure consistent node alignment within Neo4j. Next, Cypher templates map the normalized input to its corresponding intent, utilizing a schema mapping table (Table 2) that categorizes queries into attribute retrieval, multi-hop reasoning, and Boolean verification. For multi-condition queries, a filter push-down approach injects constraints directly into the Cypher WHERE clause, efficiently executing logical intersections at the database level (Francis et al., 2023). Finally, the system executes a hybrid search that merges exact string matching with contextual semantic matching via the UNION operator, significantly enhancing both retrieval coverage and precision (Gao et al., 2023).

Table 2. Representative Intent groups and Cypher mapping logic.

Intent Group	Example Intent Code	Cypher Macro Logic	Inference Goal
Entity Retrieval	FIND_FLAVOR, FIND_PROPERTY	MATCH (s)-[r:HAS_PROPERTY|HAS_FLAVOR]->(t)	Extracting original herbal properties
Formula Analysis	FIND_COMPONENTS	(f:Formula)-[:CONTAINS_HERB]->(h:Herb)	Decoding formula composition
Reverse Inference	FIND_TREATMENT	(s)-[:TREATS_DISEASE]->(d:Disease)	Finding treatments based on symptoms
Logic Verification	VERIFY_BOOLEAN	MATCH (s)-[r]->(t) WHERE t.id = $k2	Validating compatibility and interactions
Multi-layered Query	MULTI_RELATION_SEARCH	WHERE s.property = $k1 AND t.disease = $k2	Executing logic filters on multiple conditions

Upon retrieving results from the Neo4j knowledge graph, the framework executes a post-processing phase to standardize outputs for natural language generation. The retrieved JSON data is transformed into a "Fact-Sheet", a structured set of concise propositions preserving critical quantitative details such as dosage, usage instructions, and pharmacological characteristics. This Fact-Sheet functions as the explicit grounding reference, serving as the "sole source of truth" for the subsequent generation stage (Lewis et al., 2020). Following data standardization, the evidence-grounded generation process operates under rigorous parameters. Specifically, the language model's temperature is set to 0.0 to enforce deterministic outputs and reduce generative variability. Synthesized content is strictly bounded by the Fact-Sheet, successfully mitigating the risk of incorporating unverified external knowledge. Furthermore, the conversational layer integrates a strict fallback mechanism: if exact data is unavailable, it triggers a strict fallback warning and explicitly isolates closely related entities into a separate reference section to prevent clinical misinterpretation. This design constraint firmly reinforces the framework's objective as a highly regulated semantic medical retrieval system.

4 Experiments

To meet the stringent safety requirements of the healthcare domain, the framework is implemented based on the proposed decoupled 3-Agent GraphRAG Architecture. Operating on a "Safety-First" principle, the system comprises three distinct processing clusters which isolate

specific tasks to restrict the system's reliance on the free-form reasoning capabilities of Large Language Models (LLMs), thereby systematically improving the factual integrity of the retrieved medical literature. These clusters include a semantic parser dedicated to intent recognition and entity extraction, a Neo4j knowledge graph utilizing structured graph traversal as a factual anchor, and a generation module equipped with strict semantic filtering. To evaluate this ontology-grounded architecture across different processing layers, the empirical validation is formally divided into two distinct core experiments.

The first experiment focuses on validating the semantic extraction efficacy and structural robustness of the proposed Medallion data pipeline during the Silver layer processing. The foundational experimental data for this phase consists of digitized medical texts randomly sampled from the primary corpus and systematically stratified by character length (Short, Medium, and Long texts) to form a specialized testbed of 50 traditional medicinal plant species. To ensure objectivity and rigorous infrastructure auditing, this study abandons qualitative evaluation methods utilizing Large Language Models (LLM-as-a-Judge) due to their inherent indeterminacy and lack of absolute reliability in medical contexts. Instead, we established an Automated Structural and Format Audit procedure based on system log analysis. This setup employs an A/B testing methodology to benchmark two distinct approaches, directly comparing the proposed Modular Divide-and-Conquer pipeline (Decomposed 4-Module Extraction) against a Single-pass Baseline that prompts the LLM to extract all entities and relationships simultaneously. The extraction efficacy and stability are quantitatively evaluated based on technical robustness and data recovery metrics, including Complete Extraction Success Rate, Fault-Tolerance Capability (Graceful Degradation / Partial Success), Context Window Resilience, and overall Information Yield (Knowledge Graph Density).

The second experiment is designed to comprehensively evaluate the framework's end-to-end semantic retrieval capabilities and evidence-grounded generation performance. Evaluating this ontology-grounded architecture requires a tailored approach because a direct comparison with generic conversational AI or standard text-based RAG baselines may not fully reflect the system's performance, given the fundamental differences in semantic representation spaces. To establish a structurally symmetric and fair benchmark, we implemented an Advanced Sequential Pipeline (1-Agent) as the baseline. This baseline connects to the identical Neo4j knowledge graph and utilizes the same generator model but relies on a traditional, deterministic rule-based (Regex) engine for intent parsing, intentionally omitting the LLM-based NLU parser and the independent Evaluator agent. To execute this benchmark, both the proposed framework and the baseline are evaluated against a specialized test dataset comprising 240 automatically generated queries that span complex user intent scenarios, such as attribute lookups, multi-hop pathological reasoning, Boolean logic verification, and multi-constraint intersection queries. Rather than relying on superficial exact-string matching metrics, this end-to-end pipeline adopts the advanced Ragas (Retrieval Augmented Generation Assessment) methodology (Es et al., 2023), powered by the Gemini 2.5 Flash model acting as an independent judge across five core dimensions: Faithfulness, Context Recall, Context Precision, Answer Correctness, and Answer Relevancy.

Regarding the technical deployment and environment setup, all data ingested into the graph underwent a rigorous preprocessing pipeline for identity normalization and entity linking to systematically mitigate academic vocabulary ambiguities. The centralized knowledge graph was deployed using Neo4j Community Edition version 5.1, while the core natural language generation and extraction modules utilized instruction-tuned LLMs (specifically from the Gemini 2.5

family) configured with a strict temperature of 0.0 to ensure deterministic outputs and suppress generative variability. All structural graph traversals, semantic parsing workflows, and data processing pipelines were executed locally on a dedicated hardware workstation equipped with an NVIDIA GeForce RTX 3060 GPU and 16 GB of RAM to handle the multi-hop reasoning and data transformation tasks efficiently.

To rigorously validate the effectiveness and structural stability of the modular divide-and-conquer strategy at the Silver layer, an A/B testing experiment was conducted on 50 medicinal herb samples, systematically stratified by character length (Short, Medium, and Long). The evaluation utilized an automated log-based audit to detect parsing failures, truncated outputs, and schema violations, alongside a quantitative assessment of the total semantic relationships recovered. The aggregate evaluation metrics computed by the structural audit are summarized in Table 3.

Table 3. Aggregate performance of knowledge extraction over 50 stratified test samples

Evaluation Metric	Baseline (Single-pass)	Proposed (Pipeline)	Evaluation / Delta
Total Test Samples	50	50	-
Complete Success Rate	42 / 50 (84%)	49 / 50 (98%)	+14.0%
Partial Success (Fault-Tolerance)	0 / 50 (0%)	1 / 50 (2%)	+2.0% (Rescued)
Total JSON Truncation Errors	8 / 50 (16%)	0 / 50 (0%)	Eliminated
Information Yield (Relationships)	708	1335	+88.6%

The empirical results in Table 3 demonstrate that the modular Medallion architecture provides absolute structural stability and a substantial improvement over the traditional Single-pass prompting method, particularly concerning Context Window Resilience and Information Yield.

The severe performance degradation of the Baseline is evident, with 8 out of 50 files (16%) failing completely during processing. These failures exclusively occurred in the Medium and Long text strata. System log analysis revealed that all 8 failed instances returned a critical infrastructure error: JSON Truncation/Parse Error. This clearly illustrates the technical limitations of LLMs regarding the "Lazy Generation" phenomenon and Context Window overload. When forced to process dense OCR text and simultaneously generate a monolithic JSON structure containing complex chemical arrays, formulas, and medical properties, the model reaches its operational limits. Consequently, the outpsut string is abruptly truncated, missing closing structural brackets (},]), which causes the system's parsers to fail entirely, resulting in a 100% data loss for those samples.

In contrast, the proposed modular pipeline completely resolved this issue, reducing the total structural error rate to zero. By systematically decomposing the problem into four independent task modules (Identification, Chemical Composition, Clinical Properties, and Modern Pharmacology), the system ensures that the LLM's attention mechanism is maximally focused. The resulting JSON outputs from each sub-task are highly compact and safely within the model's token limits. Notably, the framework demonstrated a crucial Fault-Tolerance capability: in the single instance where a specific module encountered length issues (the Long-stratum Tắc kè sample), the pipeline successfully preserved the outputs of the remaining three modules (Partial Success), rescuing 75% of the data instead of suffering a total loss.

Furthermore, the most significant advantage of the proposed architecture is its Information Yield. By reducing the cognitive load on the LLM per prompt, the multi-agent pipeline successfully extracted 1,335 semantic relationships compared to the Baseline's 708, representing an 88.6% increase in overall Knowledge Graph Density. Once all child processes succeed, the system merges them into a complete ontological data block. This approach not only guarantees

100% robustness for the core data processing pipeline but also substantially enriches the depth and coverage of the medical ontology.

The evaluation results of the framework on the 240-query test set, utilizing the RAGAS methodology, are detailed in Table 4. To rigorously benchmark the proposed decoupled 3-Agent architecture, we establish an Advanced Sequential Pipeline (1-Agent) as the baseline. This baseline employs a comprehensive hierarchical rule-based engine (complex Regex decision trees) for intent parsing and connects to the exact same Neo4j knowledge graph, ensuring a structurally symmetric and fair comparison. It purposefully omits the LLM-based NLU parser and the independent Evaluator agent to empirically highlight the limitations of traditional deterministic algorithms in handling complex, multi-hop medical queries. By adopting the Ragas methodology powered by the Gemini 2.5 Flash model as an independent judge, both systems were evaluated across five core dimensions: Faithfulness, Context Recall, Context Precision, Answer Correctness, and Answer Relevancy.

Table 4. Comparative performance of the 3-Agent GraphRAG vs. Advanced Sequential Baseline (N=240)

Ragas Metrics	**Baseline (1-Agent Sequential)**	**Proposed (3-Agent GraphRAG)**	Delta
Faithfulness	0.7622	0.8101	+0.0479
Answer Relevancy	0.3908	0.4628	+0.0720
Context Precision	0.7833	0.7941	+0.0108
Context Recall	0.7329	0.8251	+0.0922
Answer Correctness	0.6747	0.5992	-0.0755

Regarding semantic retrieval, the proposed framework achieved a Context Recall of 0.8251, outperforming the Baseline (0.7329). This improvement exposes the "Rule Exhaustion" limitation of the Baseline's deterministic Regex engine, which severely struggles with complex multi-hop queries. In contrast, Agent 1 (LLM Intent Analysis) deeply comprehends semantic intersections, effectively decomposing complex queries into multi-layered Cypher traversals to ensure higher coverage of relevant graph nodes.

In terms of hallucination mitigation, the 3-Agent system achieved a Faithfulness score of 0.8101 compared to the Baseline's 0.7622. While the Baseline relies on an algorithmic post-validator that is blind to contextual medical logic, Agent 3 (Evaluator) provides a rigorous semantic guardrail. By strictly cross-validating responses against structured Fact-Sheets, it successfully suppresses generative hallucinations. Additionally, semantic filtering improved Answer Relevancy (0.4628 vs. 0.3908) by systematically discarding peripheral relationships that cause verbose, off-topic responses.

A seemingly counterintuitive observation is the Baseline's higher Answer Correctness score (0.6747 vs. 0.5992). However, this discrepancy is a structural artifact of the evaluation metric. The dataset's ground truths consist of brief, mechanically generated templates, which the 1-Agent Baseline trivially replicates, artificially inflating its score. In contrast, the 3-Agent framework is designed to produce comprehensive, well-structured responses that mirror the depth of clinical medical literature. While this more detailed articulation, which provides broader context rather than simple template-matching, results in a lower algorithmic correctness score, it offers vastly superior interpretability and pedagogical value for supporting medical professionals in evidence-based decision-making.

Finally, for the 'CONTRAINDICATED_FOR' (Contraindication) relationship, AI-driven reasoning was deliberately suppressed. To ensure clinical safety, the framework only retrieves contraindications explicitly documented in the authoritative medical literature, prioritizing structural evidence over generative assumptions to mitigate severe medical hallucinations.

5 Conclusion

This study proposes an ontology-grounded semantic retrieval framework for Traditional Vietnamese Medicine (TVM), utilizing a decoupled 3-Agent architecture. By systematically translating unstructured historical texts into an ontology-driven Knowledge Graph via a Medallion pipeline, the system establishes a highly structured semantic space. This explicitly models complex therapeutic relationships and acts as a strict factual anchor. Empirical results demonstrate that our architecture significantly outperforms a deterministic sequential baseline in multi-hop reasoning (achieving a Context Recall of 0.8251) and substantially mitigates hallucination risks (Faithfulness of 0.8101).

Despite these promising results, certain limitations exist. The foundational Knowledge Graph is derived from historical texts without comprehensive modern clinical validation. Additionally, while Large Language Models (LLMs) handle dynamic intent analysis, the reliance on predefined Cypher templates restricts retrieval flexibility for out-of-schema queries. Furthermore, to prioritize clinical safety, generative AI reasoning is strictly prohibited for undocumented contraindications.

Future research will focus on integrating a 'Clinical Expert-in-the-Loop' module for manual validation and exploring neuro-symbolic parsing approaches to dynamically handle semantic ambiguities. Ultimately, deploying the framework in real-world scenarios will further refine its capability to safely assist evidence-based decision-making in traditional medicine.

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